

The Role of AI in HCI Education

Per Ola Kristensson
Department of Engineering
University of Cambridge
Cambridge, United Kingdom
pok21@cam.ac.uk

Antti Oulasvirta
Aalto University
Helsinki, Finland
antti.oulasvirta@aalto.fi

Elizabeth F Churchill
Mohamed bin Zayed University of
Artificial Intelligence
Abu Dhabi, United Arab Emirates
churchill@acm.org

Malak Sadek
Centre for Human-Inspired Artificial
Intelligence
University of Cambridge
Cambridge, United Kingdom
mfzas2@cam.ac.uk

Lin-Ping Yuan
Department of Computer Science and
Engineering
The Hong Kong University of Science
and Technology
Hong Kong SAR, China
yuanlp@cse.ust.hk

Kasper Hornbæk
Department of Computer Science
University of Copenhagen
Copenhagen, Denmark
kash@di.ku.dk

Albrecht Schmidt
LMU Munich
Munich, Germany
albrecht.schmidt@ifi.lmu.de

Abstract

While AI has a long history in HCI, AI is now readily accessible to everyone and providing a level of performance that makes the use of AI tractable across a wide range of applications. As such, HCI needs to seriously consider the role of AI in HCI education. This meet-up serves as a catalyst to create a community within HCI, discussing how we teach HCI in the era of AI from four perspectives: 1) theory; (2) practice; (3) assessment; and (4) an inclusive community.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**.

ACM Reference Format:

Per Ola Kristensson, Elizabeth F Churchill, Kasper Hornbæk, Antti Oulasvirta, Malak Sadek, Albrecht Schmidt, and Lin-Ping Yuan. 2026. The Role of AI in HCI Education. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3772363.3778885>

1 Introduction

AI has a long rich history in HCI and HCI researchers have created new ways of interaction, such as auto-correct [4], interfaces that adapt to users' capabilities [3], and ways of enabling users to interact with their own body through touch [5]. In addition, the HCI field has studied the impact of speech recognition [7] and

devised new approaches to work with AI, such as Wizard-of-Oz protocols [1], mixed-initiative interaction [6], and interactive machine learning [2].

However, it is also clear that AI is now at a stage where it is not only readily accessible to everyone but also at a level of performance where a wide range of applications of AI are suddenly feasible. These developments raise the bar for HCI educators—are established HCI theories and methods still applicable in an era of ubiquitous AI, and do we need to change the way we teach and assess HCI?

2 Meet-up Goal

The goal of this meet-up is to serve as a catalyst for a community-wide discussion on how we ought to consider AI in HCI education going forward. Specifically, we aim to have discussions on the following four topics:

Theory. *What are the roles of HCI theories in an AI era?* HCI benefits from a range of established theories and approaches, ranging from KLM-GOMS, Fitts' law, and information scent, to contextual inquiry and activity theory. How should we teach HCI theory in light of advances in AI? What are the aspects of HCI theory that contribute to AI-relevant HCI teaching and are there any pressing gaps in theory we need to fill—either by creating new theory or by adopting theories for neighboring fields, such as human factors, experimental psychology, behavioral economics, computational neuroscience, or machine learning?

Practice. *How should we teach HCI methods and processes in the AI era?* The current struggles large organizations face in incorporating AI makes the case for taking HCI seriously more stronger than ever. Successfully utilizing AI necessitates understanding organizations, role structures, users, needs, wants, values, tasks, and processes—all areas where HCI has adopted and developed successful methods ranging from hierarchical task analysis to analysis of appropriation



This work is licensed under a Creative Commons Attribution 4.0 International License. *CHI EA '26, Barcelona, Spain*

© 2026 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2281-3/26/04

<https://doi.org/10.1145/3772363.3778885>

patterns. At the same time, traditional HCI teaching does not focus on successful AI adoption, nor does it tackle actionable design methods for designing and building usable AI systems. How do we incorporate recent HCI research and adopt developments from neighboring fields, such as machine learning, in our teaching in effective and efficient ways?

Assessment. *How do we assess HCI knowledge, insights, and skills in the era of AI?* Large language models provide students with rapid design, programming, and report writing capabilities. However, students also need to work with the material and practice to gain insights and to be able to reflect on their practice. On the other hand, advancement in AI provides the potential to accelerate HCI projects and thus enable students to learn from a richer set of experiences.

An inclusive community. *How do we best include nascent developments in nearby fields and build bridges regarding AI teaching across disciplines?* HCI has always relied on a plurality of disciplines, such as psychology, computer science, cognitive science, design, human factors, engineering, and the humanities. AI raises the stakes as there are serious issues of governance, such as fairness and equity, non-discrimination, accountability, explainability and accountability, and security and privacy. Tackling these issues requires HCI teaching to synthesize insights from many fields, for example, technical advances in explainability frequently arises in machine learning. Further, neighboring fields may benefit from HCI insights. How do we best foster a cross-disciplinary focus in our HCI teaching?

3 Activities

As the major aim is to bring together HCI educators we want to make the meet-up as interactive as possible. The 90-minute meet-up will be structured into three parts.

The first 40-minute part will be a participant-active panel discussion with a moderator. We will through a moderator ask the panel to ponder on the role of AI in HCI education by asking the panel to reflect on four topics: (1) theory; (2) practice; (3) assessment; and (4) an inclusive community. The moderator and the panel will be selected from the meet-up organizers. To make the most of the meet-up time slot, the moderator will briefly introduce the panelists. To engage all participants, we will leverage an experienced moderator who will actively engage participants in the conversation.

Recognizing that HCI educators may be particularly interested in specific aspects, and to encourage networking, the second 40-minute part will consist of clustered discussions. We will cluster the room into the four above topics and the organizers will be distributed across these clustered discussions.

The final 10-minute part will be an all-room discussion. The organizers will engage in a community discussion on next steps, such as a mailing list, a social media group, and further meet-up and conference activity.

4 Organizers

The organizers are all HCI educators active in the SIGCHI community.

Per Ola Kristensson is the lead organizer. He is a Professor of Interactive Systems Engineering at the Department of Engineering, University of Cambridge, and a co-founder and co-director of the

university-wide Centre for Human-Inspired Artificial Intelligence. He is a co-author of the recently published book *Introduction to Human-Computer Interaction* (Oxford University Press), which is a comprehensive Open Access HCI textbook (864 pages, 46 chapters). He has taught HCI for approximately 20 years and recently set up a new MPhil program in human-inspired AI. He served as a CHI 2023 Technical Program Co-Chair and is an associate editor of ACM Transactions on Computer-Human Interaction and ACM Transactions on Interactive Intelligent Systems.

Elizabeth Churchill is Department Chair and Professor of Human-Computer Interaction at the Mohamed bin Zayed University of Artificial Intelligence. She has held numerous roles at ACM and ACM SIGCHI. She is currently an editor of *Interactions*, a member of the CHI Academy, an ACM Fellow, and received the ACM SIGCHI Lifetime Service Award in 2023 and the ACM SIGCHI Lifetime Practice Award in 2024.

Kasper Hornbaek is a Professor in Human-Centered Computing at the Department of Computer Science, University of Copenhagen. In addition to being a co-author of the aforementioned book *Introduction to Human-Computer Interaction* and teaching HCI for over twenty years, he is editor-in-chief of ACM Transactions on Computer-Human Interaction and a member of the CHI Academy.

Antti Oulasvirta is a Professor of Electrical Engineering and leads the User Interfaces research group at Aalto University and the Interactive AI research program at the Finnish Center for AI. In addition to being a co-author of the aforementioned book *Introduction to Human-Computer Interaction* and teaching HCI for over twenty years, he is an associate editor of ACM Transactions on Computer-Human Interaction and a member of the CHI Academy.

Malak Sadek is a Teaching Associate at the Centre for Human-Inspired Artificial Intelligence at the University of Cambridge. She received her PhD in Design Engineering in 2025 from Imperial College London. She is overseeing a new MPhil programme in human-inspired AI and designing and teaching modules relating to HCI and conversational user interfaces.

Albrecht Schmidt is a Professor at the Ludwig Maximilian University of Munich. He has been teaching HCI for over 25 years. He is a member of the CHI Academy and an ACM Fellow.

Linping Yuan is a Research Assistant Professor at the Department of Computer Science and Engineering in Hong Kong University of Science and Technology. She received her PhD in 2024. She has been teaching since 2021 and her research interests are in the intersection of visualization, extended reality, and human-computer interaction.

5 Community

We anticipate a very high interest in this topic across SIGCHI and the entire HCI community. The primary target audience does not only include established HCI educators but also practitioners, researchers, and emerging HCI educators. In addition, we will endeavor to include both undergraduate and graduate students to represent student perspectives in this meet-up.

References

- [1] Nils Dahlbäck, Arne Jönsson, and Lars Ahrenberg. 1993. Wizard of Oz studies: why and how. In *Proceedings of the 1st international conference on Intelligent user interfaces*. 193–200.

- [2] Jerry Alan Fails and Dan R Olsen Jr. 2003. Interactive machine learning. In *Proceedings of the 8th international conference on Intelligent user interfaces*. 39–45.
- [3] Krzysztof Gajos and Daniel S Weld. 2004. SUPPLE: automatically generating user interfaces. In *Proceedings of the 9th international conference on Intelligent user interfaces*. 93–100.
- [4] Joshua Goodman, Gina Venolia, Keith Steury, and Chauncey Parker. 2002. Language modeling for soft keyboards. In *Proceedings of the 7th international conference on Intelligent user interfaces*. 194–195.
- [5] Chris Harrison, Desney Tan, and Dan Morris. 2010. Skinput: appropriating the body as an input surface. In *Proceedings of the SIGCHI conference on human factors in computing systems*. 453–462.
- [6] Eric Horvitz. 1999. Principles of mixed-initiative user interfaces. In *Proceedings of the SIGCHI conference on Human Factors in Computing Systems*. 159–166.
- [7] Clare-Marie Karat, Christine Halverson, Daniel Horn, and John Karat. 1999. Patterns of entry and correction in large vocabulary continuous speech recognition systems. In *Proceedings of the SIGCHI conference on Human Factors in Computing Systems*. 568–575.